AI-Assisted Coding for Educators

AI-assisted coding

AI coding tools

Human thinking + AI collaboration

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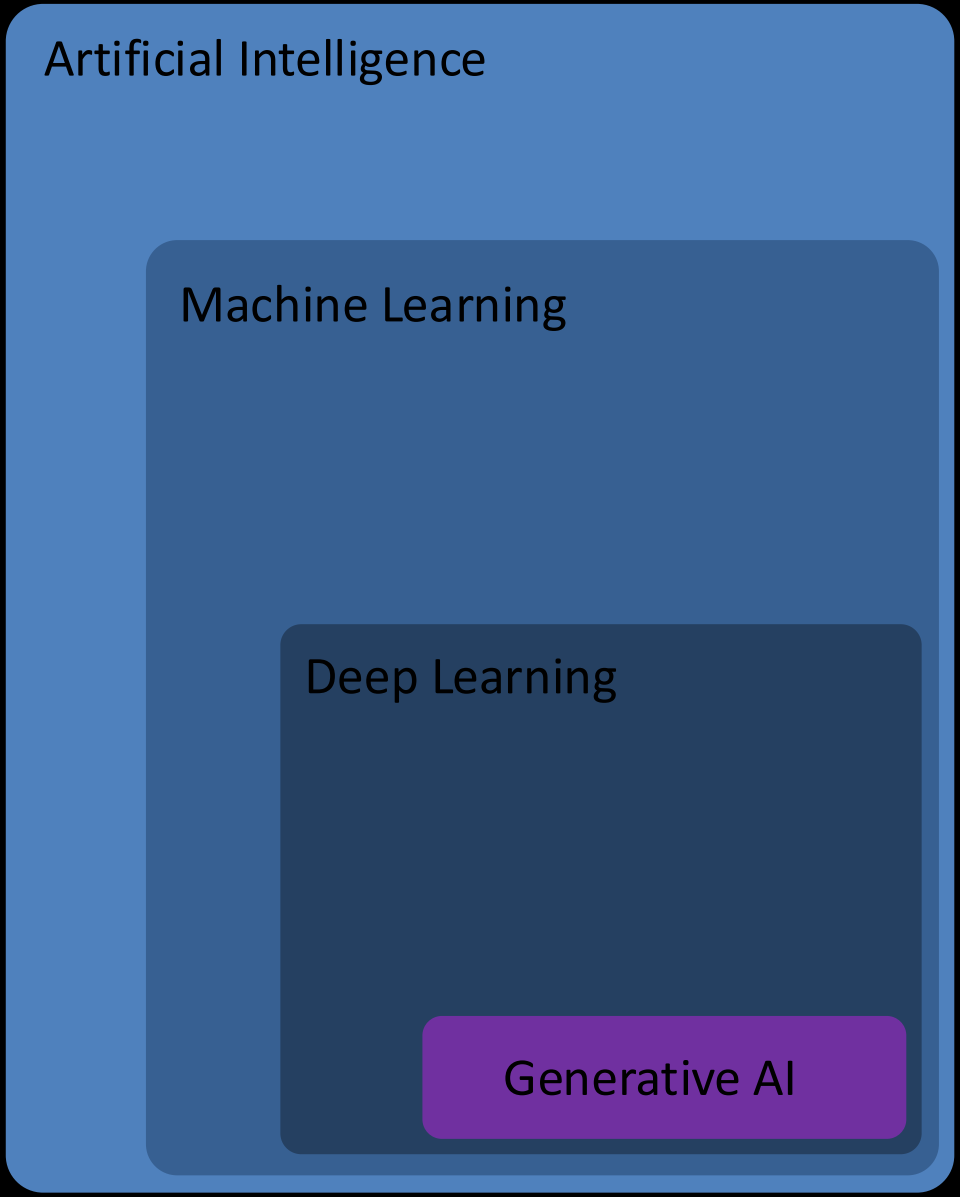
Agenda

- Generative AI
- AI assisted Coding
  - Tools
  - Risks
- Takeaways

GenAI



1956

Artificial Intelligence

Intelligence demonstrated by machines

1997

Machine Learning

Learn from data

2017

Deep Learning

Model after the human brain (Neural Networks)

2021

Generative AI

Create new written, visual, and auditory content

Generative AI

The best thing about AI is its ability to …

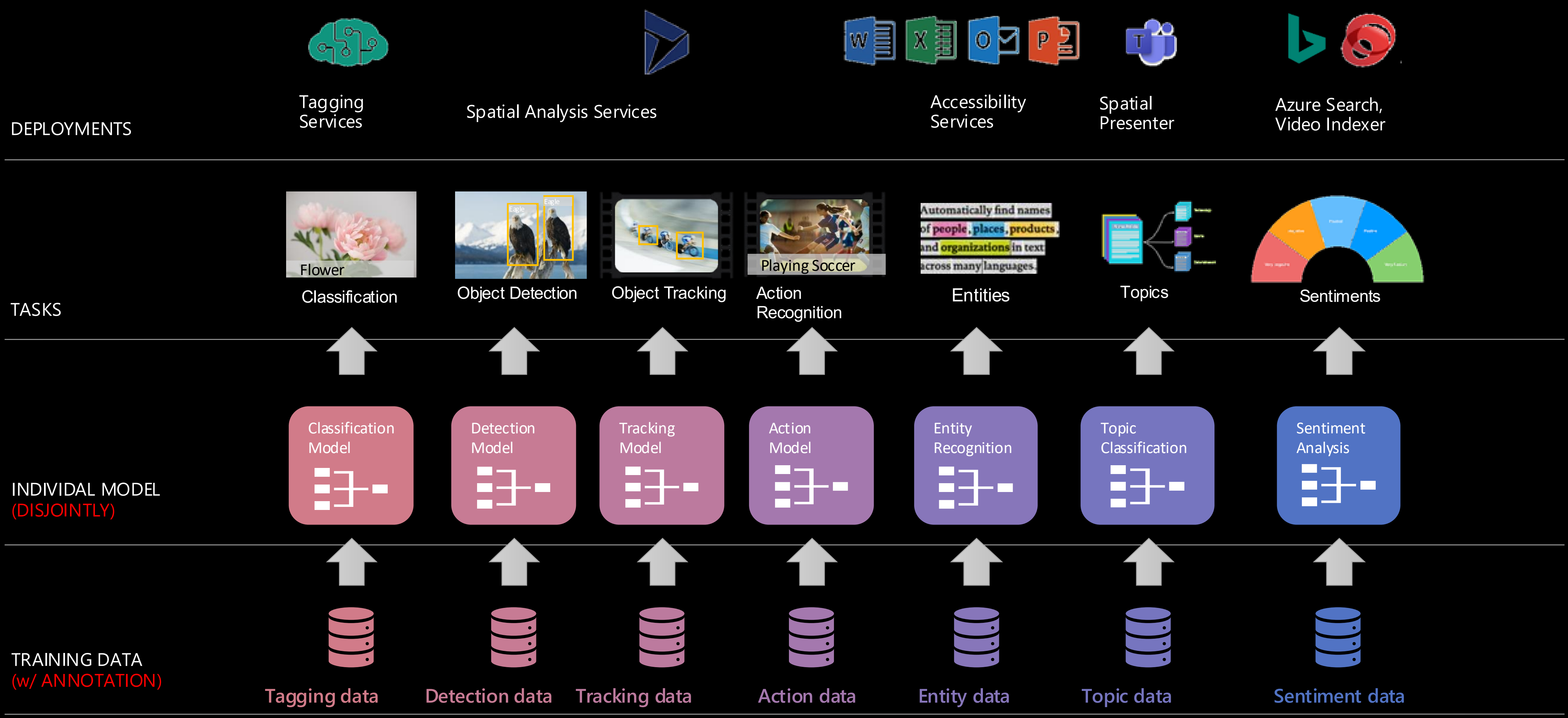
Adapt = 30%

Process = 22%

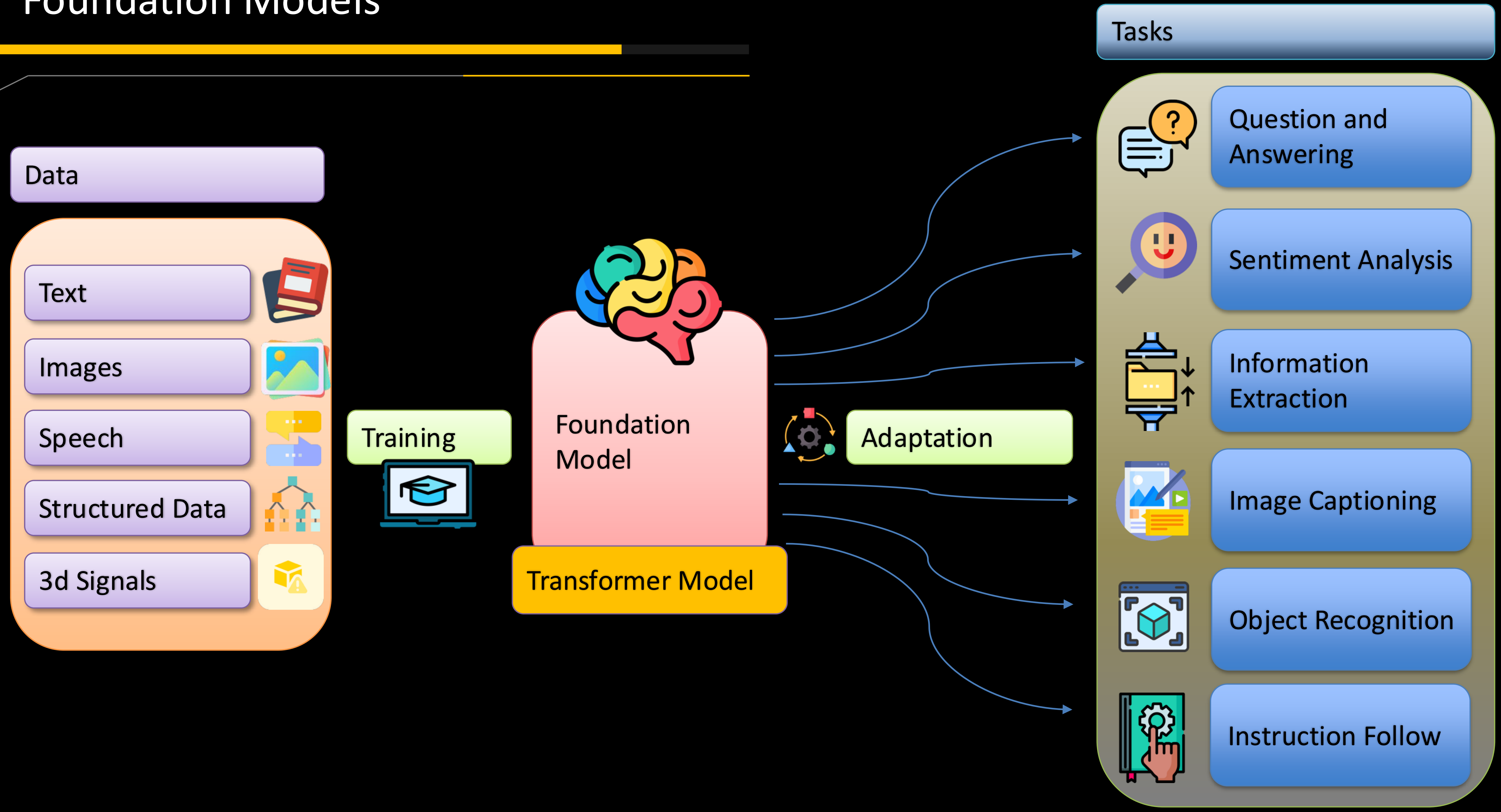
Analyze = 7%

Traditional model development

High cost & slow deployment - Each service is trained disjointly



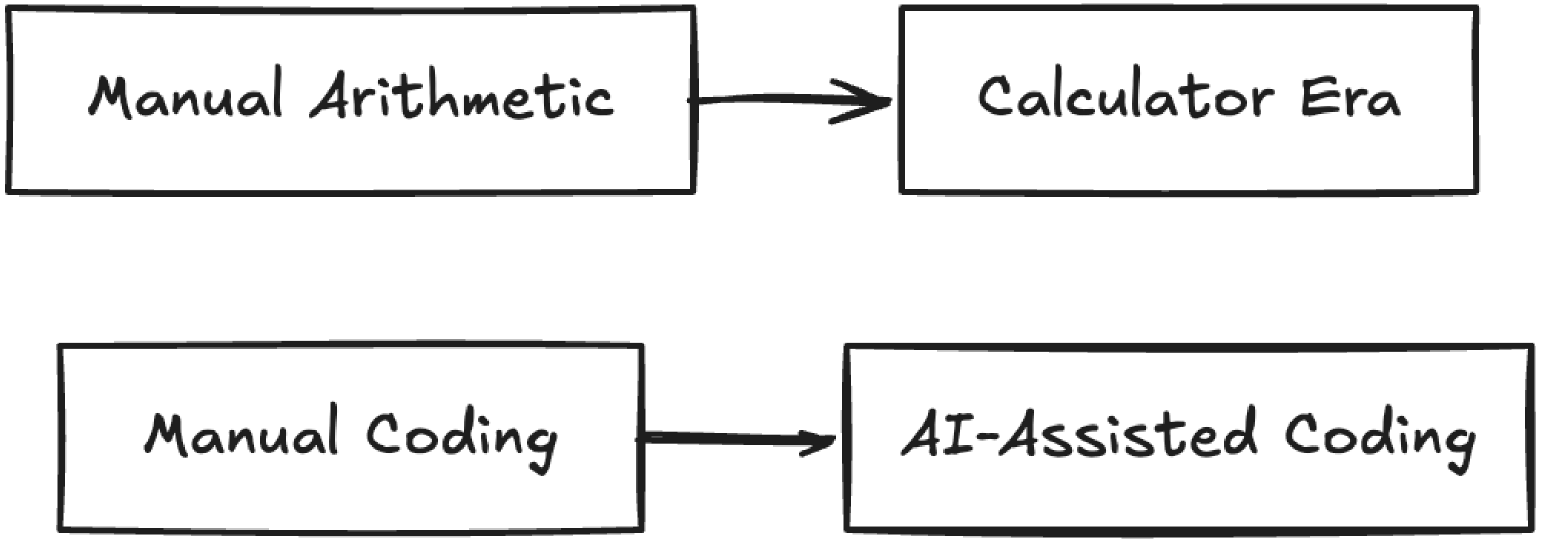
Foundation Models



Why This Matters

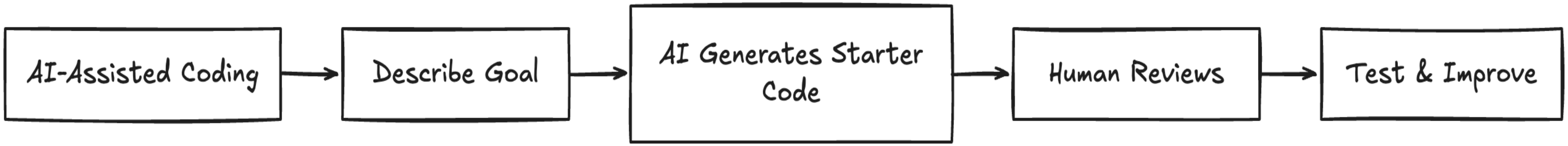
- AI is changing software development
- Students will work with AI professionally
- Goal: teach thinking, not memorization

Calculator Analogy



- Calculators changed math education
- AI coding tools change (programming) education
- Humans still provide reasoning and verification

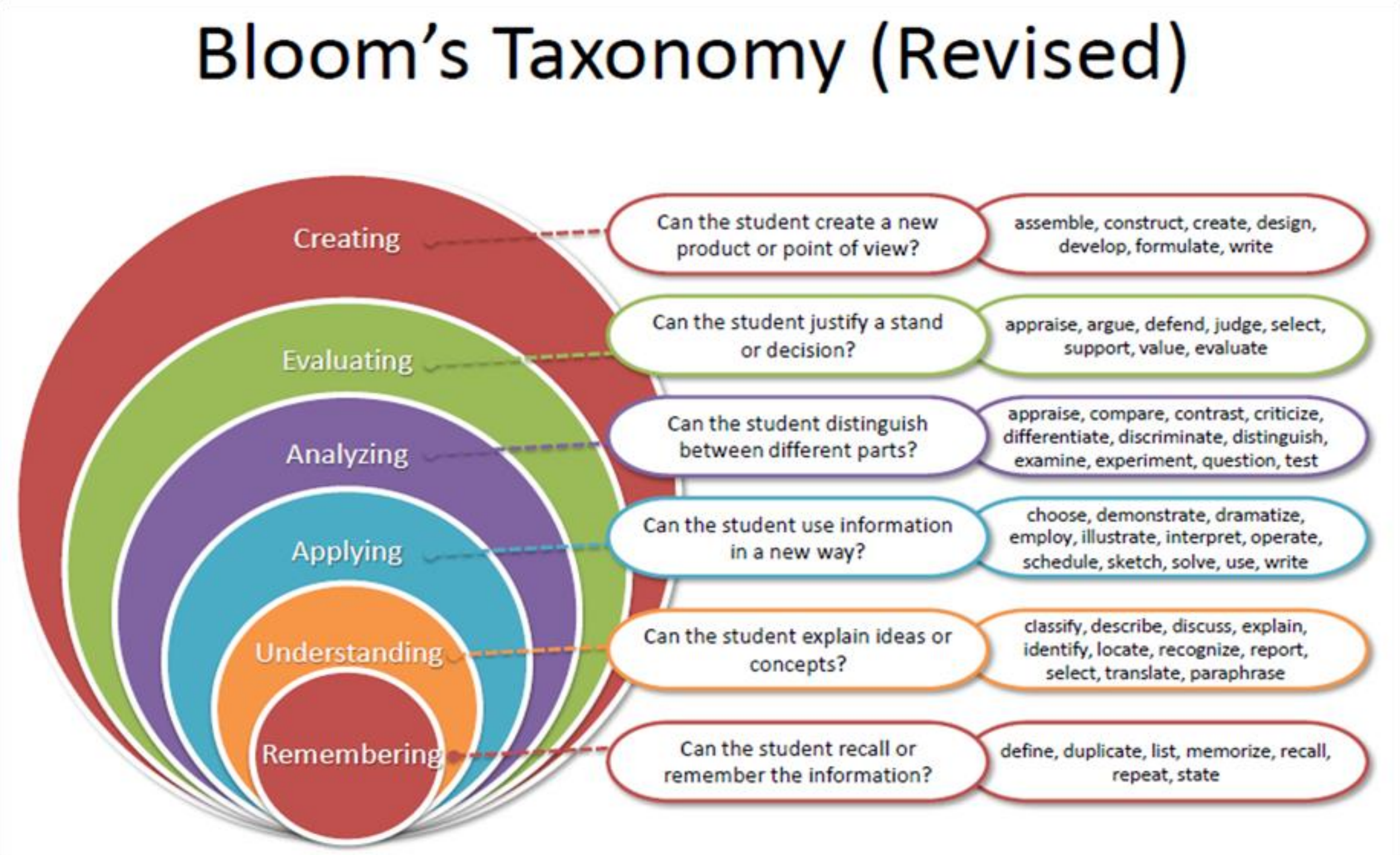
Traditional vs AI-Assisted Coding



- Before AI: syntax, debugging, boilerplate
- With AI: design, testing, evaluation
- Shift toward higher-order thinking

Design

- Learning objectives
- Select instructional strategies
- Select assessment methods

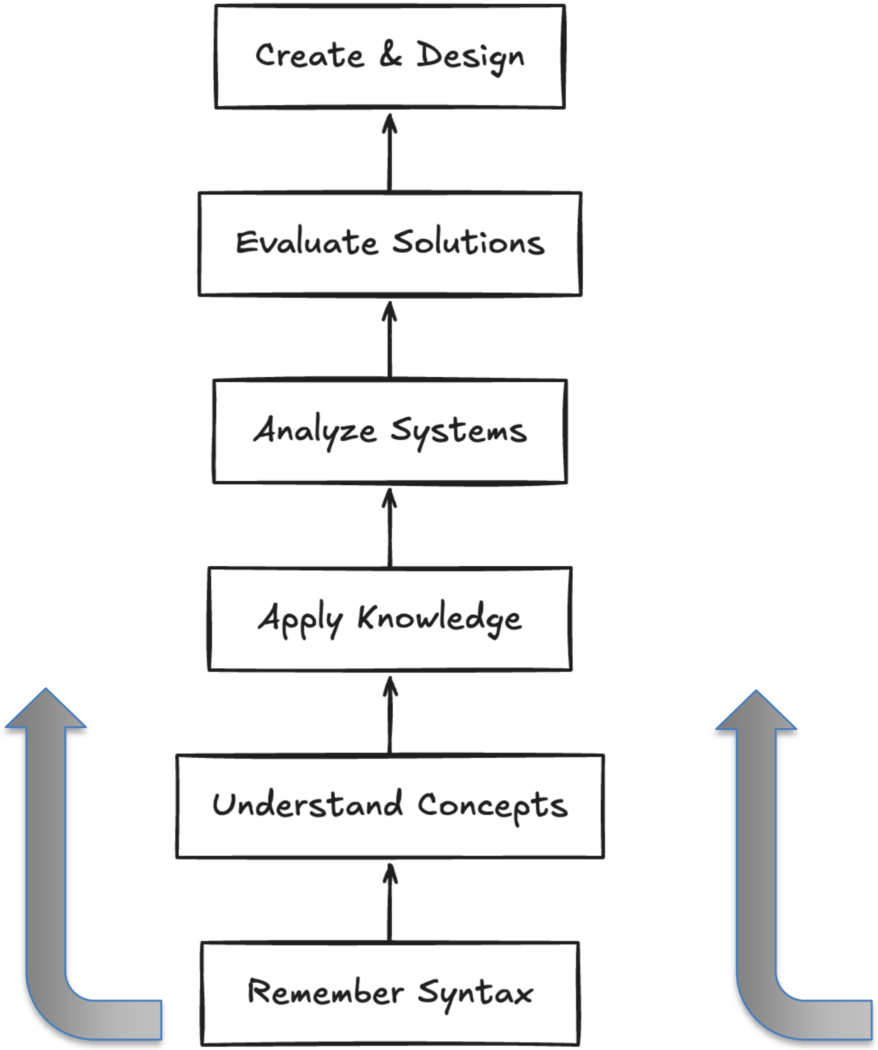


Bloom's Taxonomy Shift

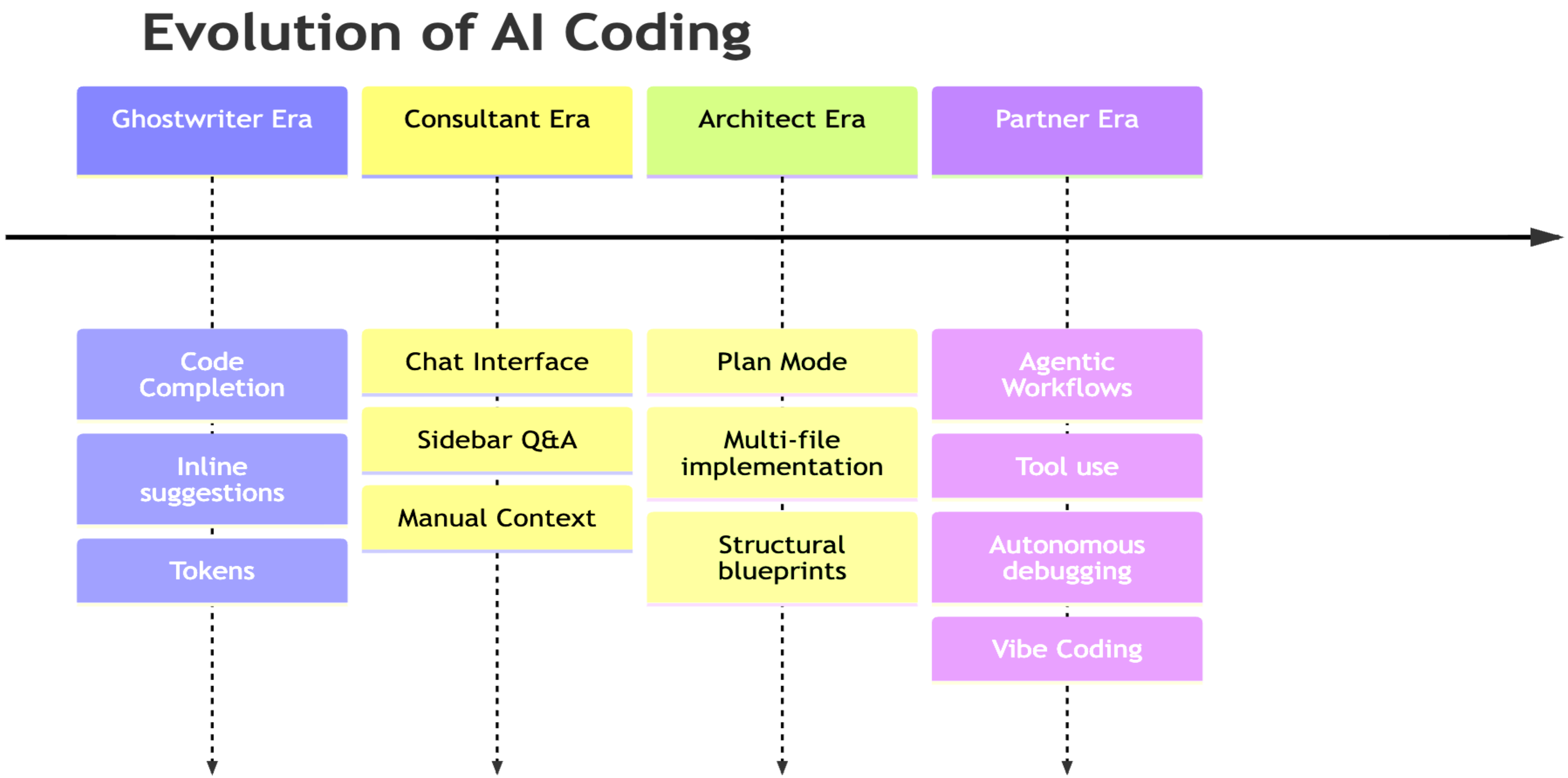
- Less time on syntax
- More time on analysis and design
- AI can elevate classroom thinking

SAMR Model (Enhancement-Transformation)

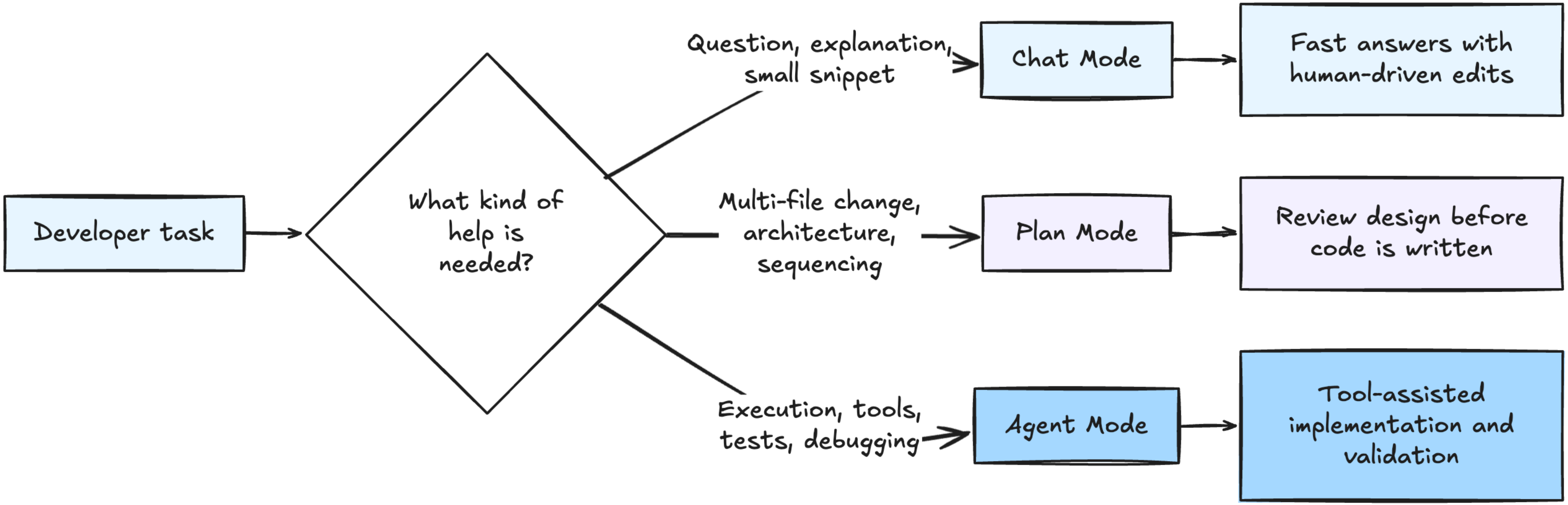
- Substitution
- Augmentation
- Modification
- Redefinition



Tooling transformation



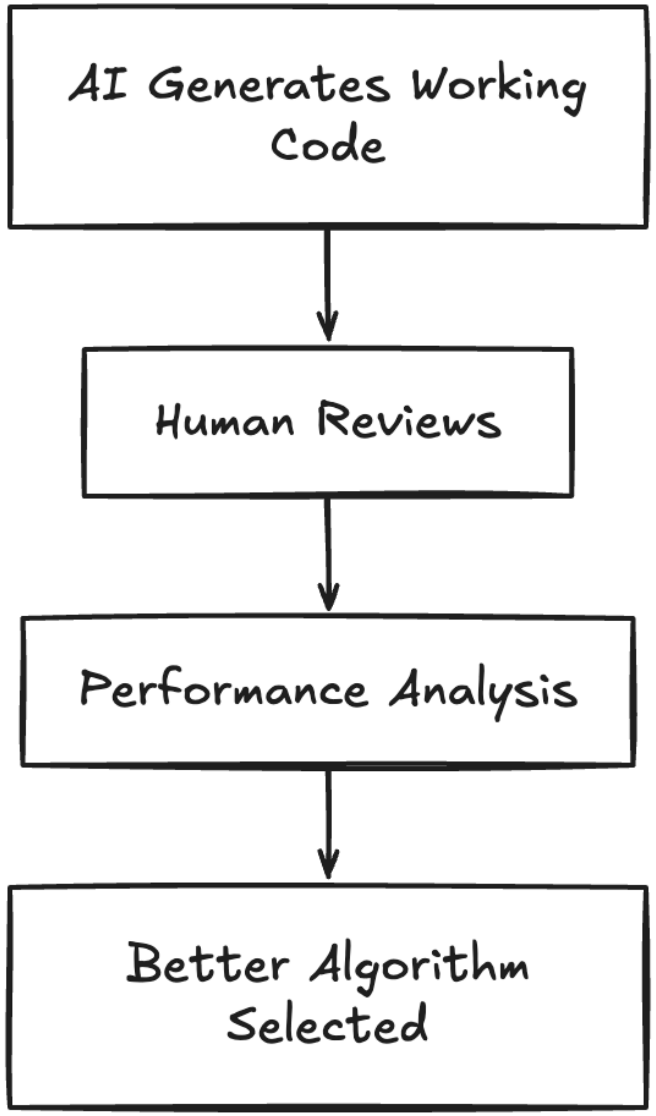
IDE* Modes for AI-Assisted Coding



*Integrated Development Environment (IDE) – e.g.: Visual Studio Code

Prime Numbers Example

`Write a script to calculate primes less than N`

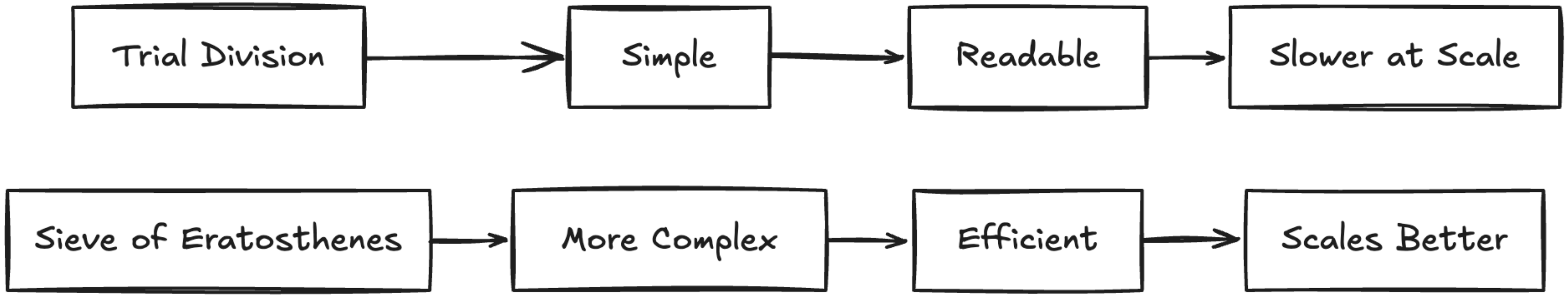


- AI often suggests simpler solutions first
- Humans improve scalability
- Correct ≠ optimal

Trial Division

- Readable and beginner friendly
- Time complexity: O(n√n)
- Slows dramatically for large inputs

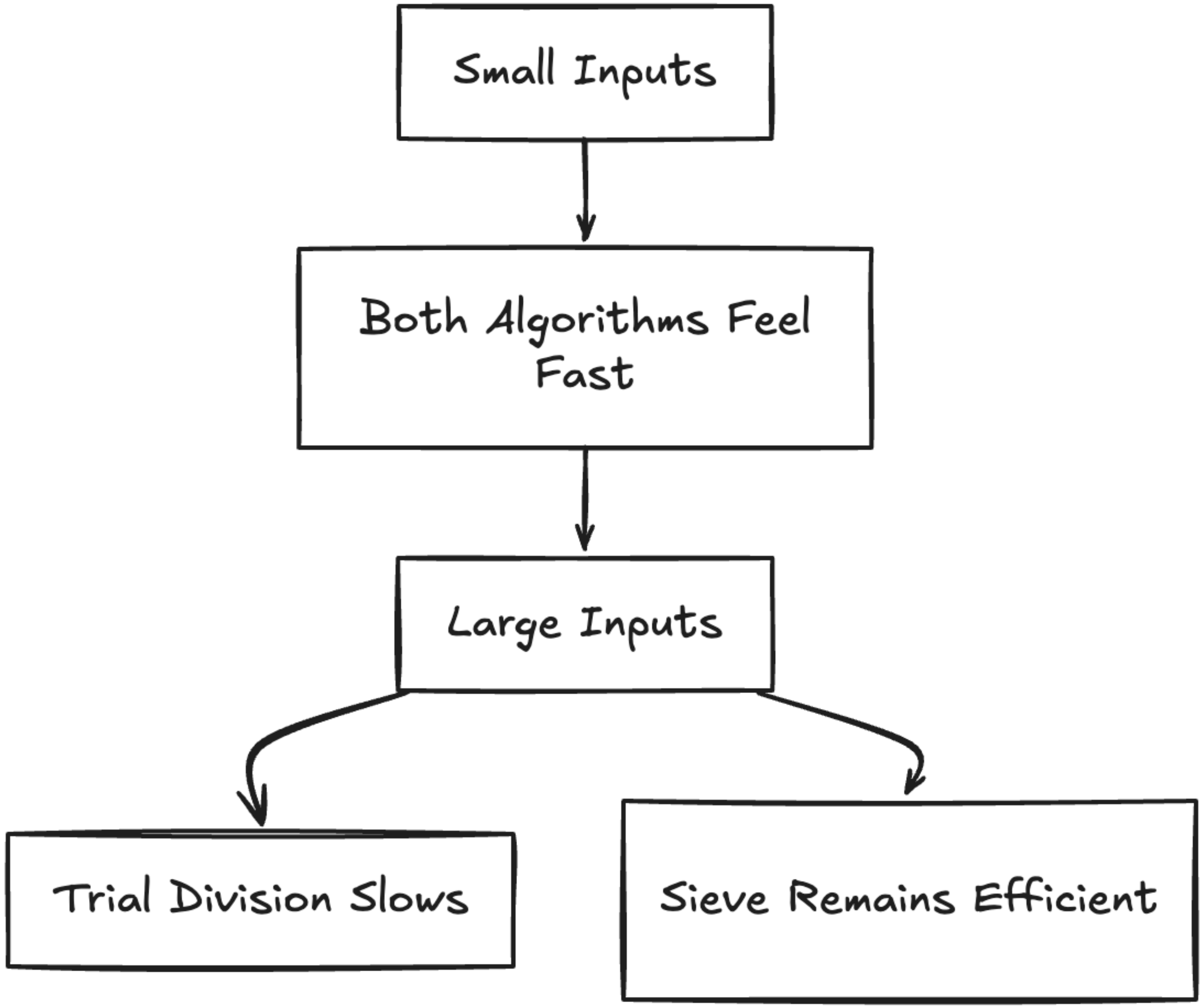
```
def is_prime(num: int) -> bool:
"""Return True if num is prime, using trial division."""
    if num < 2:
        return False
    if num == 2:
        return True
    if num % 2 == 0:
        return False
```



Sieve of Eratosthenes

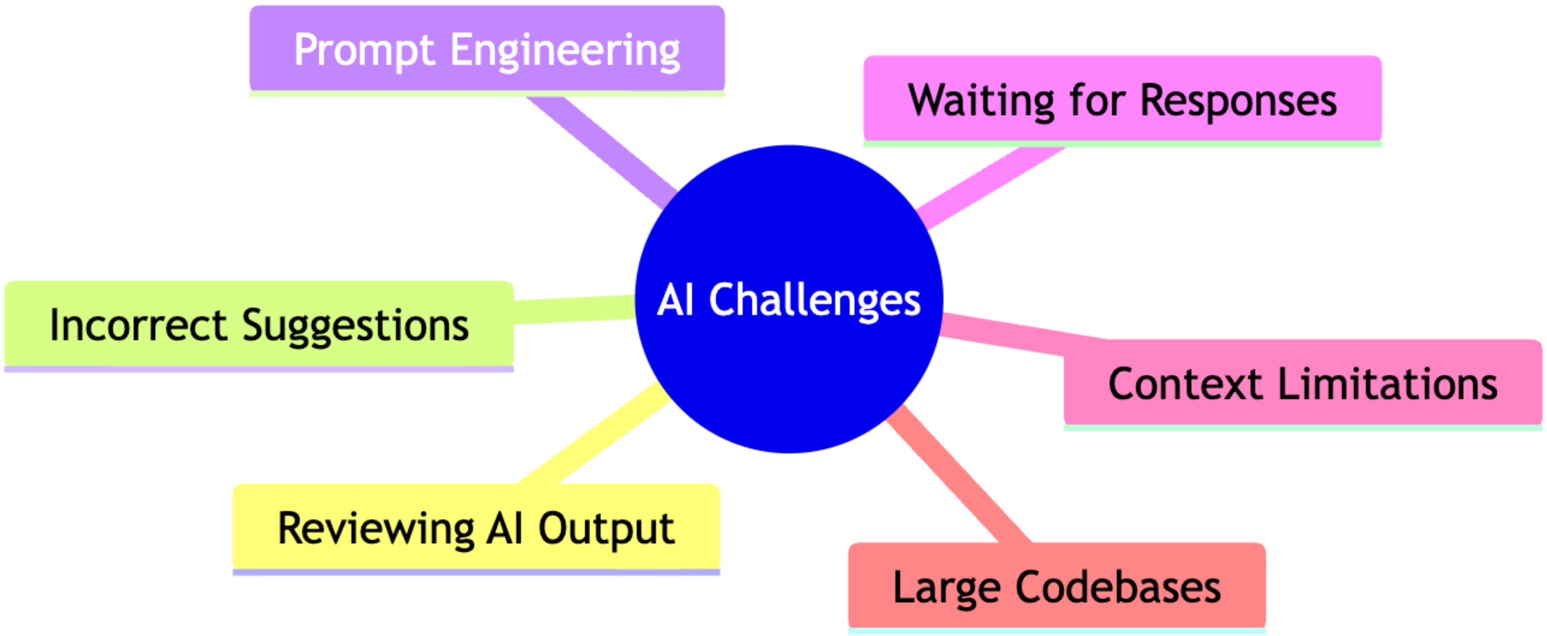
- More advanced algorithm
- Time complexity: O(n log log n)
- Much faster for large inputs

```
def primes_below(n: int) -> list[int]:
    """Return a list of all primes strictly less than n."""
    if n < 3:
        return []

    # is_prime[i] will be False once i is known to be composite.
    is_prime = [True] * n
    is_prime[0] = is_prime[1] = False

    for num in range(2, math.isqrt(n - 1) + 1):
        if is_prime[num]:
            # Mark all multiples of num (starting at num*num) as composite.
            for multiple in range(num * num, n, num):
                is_prime[multiple] = False

    return [num for num, prime in enumerate(is_prime) if prime]
```



Risks

Research Findings (METR Study)



https://metr.org/blog/2025-07-10-early-2025-ai-experienced-os-dev-study/

Research Findings (METR Study)

- Experienced developers used AI
- Tasks sometimes took 19% longer
- AI helps, but oversight matters

https://metr.org/blog/2025-07-10-early-2025-ai-experienced-os-dev-study/

Risk mitigation plan for AI-assisted Coding in Education

Key Concerns

Blind trust

Prevent shortcut learning

Demystify the tool

Safeguards

Verify AI code & understand it before use

Use manual coding & AI to build foundational skills

Teach how AI coding tools work

Risk mitigation plan for AI-assisted Coding in Education

Key Concerns

Skills gap

No human-in-the loop

Safeguards

Foster critical thinking & peer reviews to reduce

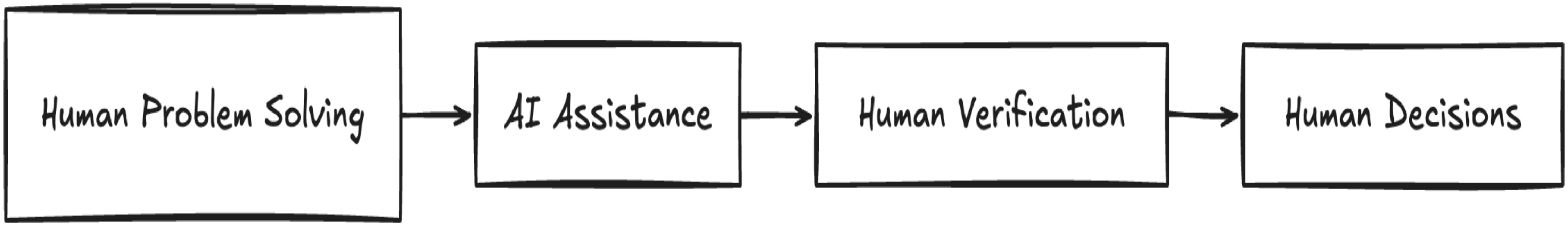
overconfidence in unverified outputs

Mandate code reviews & discussions on AI-generated code

Development with AI

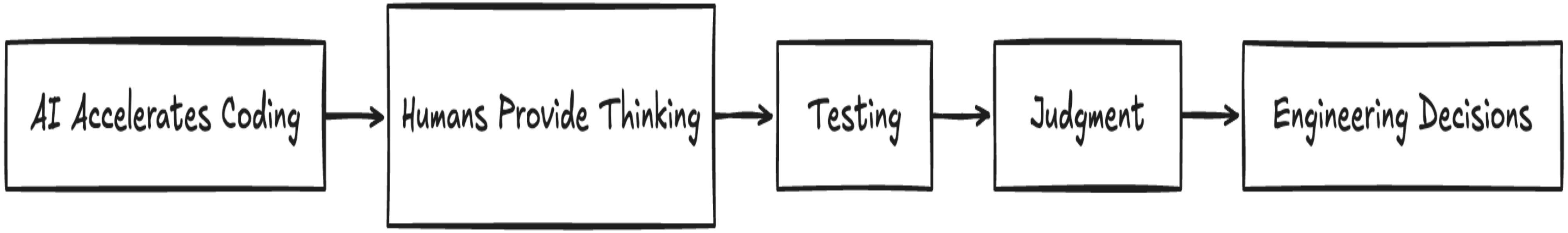
- AI generates starter code
- Students review and modify
- Human judgment remains essential

Human Thinking vs AI Suggestions



- AI accelerates implementation
- Humans optimize and evaluate
- Engineering requires tradeoffs

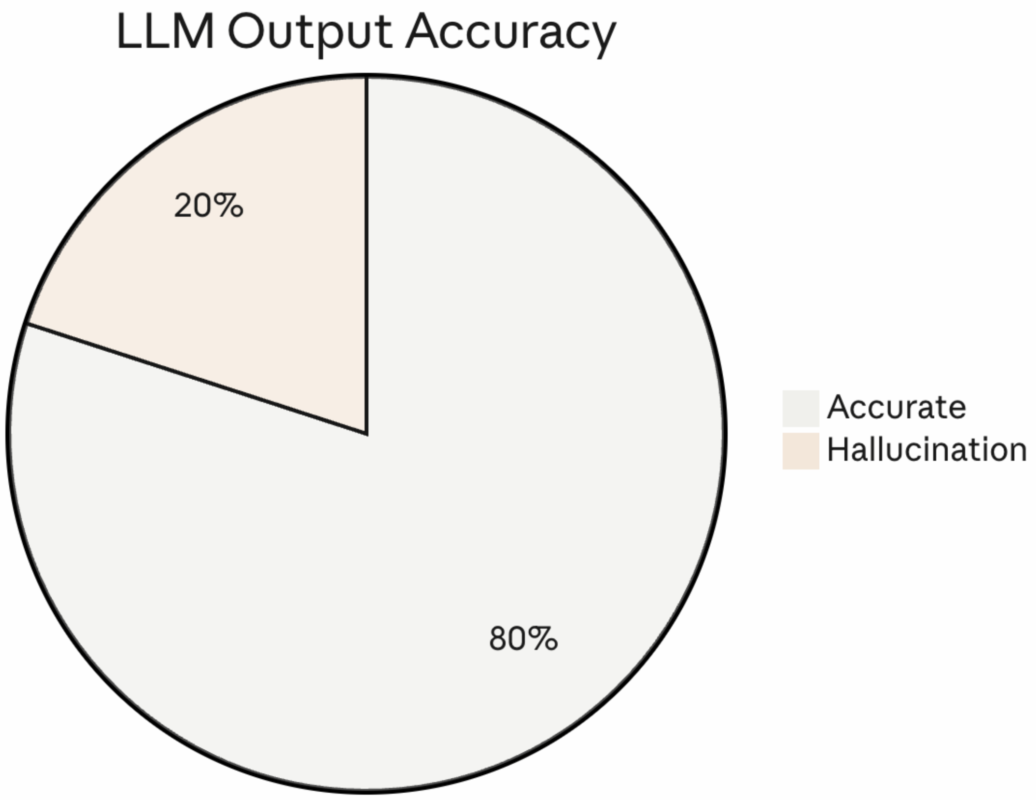
Key Message for Educators



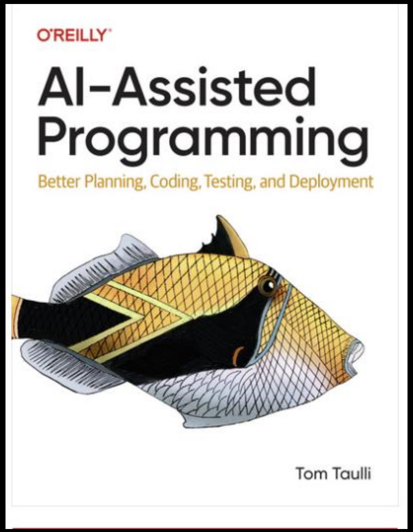
- AI does not replace thinking
- AI reduces mechanical friction
- Teach students to think critically with AI

Discussion Questions

- What should AI automate?
- What thinking should humans always do?
- How do we teach responsible AI use?



Resources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Better Planning, Coding, Testing, and Deployment
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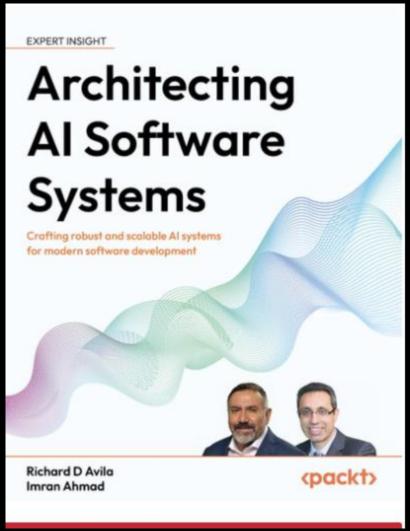
EXPERT INSIGHT
Architecting AI Software Systems
Crafting robust and scalable AI systems for modern software development
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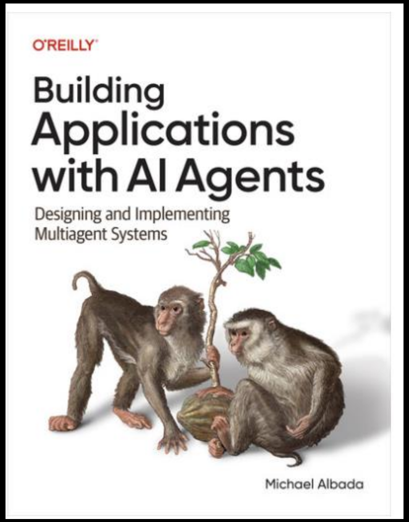
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Designing and Implementing Multiagent Systems
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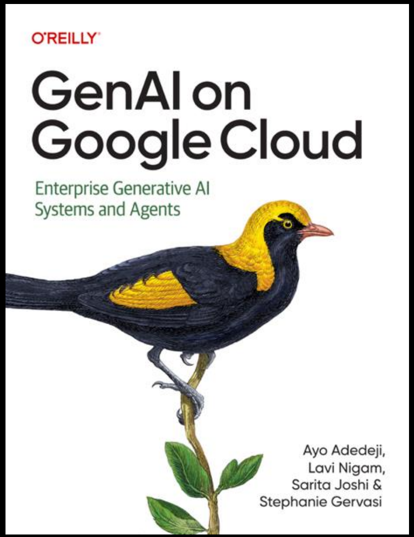
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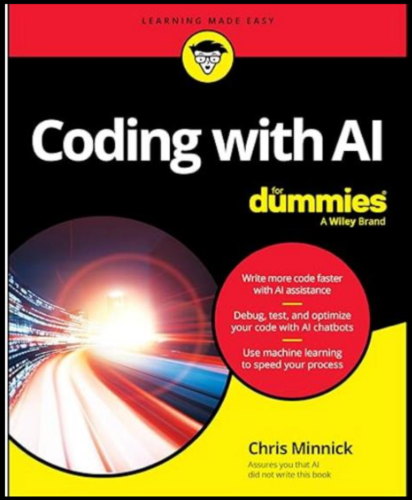
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